

Exam Date & Time: 29-Nov-2023 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. EXAMINATIONS - NOVEMBER / DECEMBER 2023

SUBJECT: MAT 1171/ MAT_1171 - ENGINEERING MATHEMATICS - I

ENGINEERING MATHEMATICS - I [MAT 1171]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

1A) The table below gives the values of $\tan x$, where x is in radians

x	0.10	0.15	0.20	0.25	0.30
$\tan x$	0.1003	0.1511	0.2027	0.2553	0.3093

(4)

Using Newton's backward difference formula, find $\tan(0.26)$.

1B) Using Newton-Raphson method, find the real root of the equation

$$\cos x = 3x - 1$$

(3)

where x is in radians, correct to three decimal places by taking $x_0 = 0.5$.1C) Using the following data, find $\frac{dy}{dx}$ at $x = 0$.

x	0	0.1	0.2	0.3	0.4
y	1.0	0.9975	0.9900	0.9776	0.8604

(3)

2A) $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$ with $y(0) = 1$ Using Runge Kutta method of order 4, find $y(0.2)$, given with $y(0) = 1$ by taking $h=0.2$. Correct to 4 decimal places.

(4)

2B) Using modified Euler's method, find

$$y(1.4), \text{ given that } \frac{dy}{dx} = 1 + \frac{y}{x} \text{ with } y(1) = 2 \text{ by taking } h = 0.2.$$

(3)

Carry out 2 iterations in each step and correct to 4 decimal places.

2C) Using Simpson's $\frac{1}{3}$ -rule, find an approximate value for the integral

$$\int_1^2 e^{\frac{-x}{2}} dx.$$

(3)

Take $h = 0.25$ and correct to 4 decimal places.3A) Using Gram-Schmidt orthogonalization process find an orthonormal basis of $\mathbb{R}^3$ from the set of vectors $\{(1,1,2), (1, -1,2), (1,0,1)\}$.

(4)

3B) (3)

$$\text{Solve } (5x^4 + 3x^2y^2 - 2xy^3)dx + (2x^3y - 3x^2y^2 - 5y^4)dy = 0$$

3C) Solve $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = e^x \sin x$ (3)

4A) Prove that in a vector space V over a field F , every maximal linearly independent set of vectors forms a basis for V . (4)

4B) Solve $x^2y'' - xy' + 2y = x \sin(\log x)$ (3)

4C) Solve by Gauss Elimination Method:

$$\begin{aligned} 3x_1 - 4x_2 + 5x_3 &= -1 \\ -3x_1 + 2x_2 + x_3 &= 1 \\ 6x_1 + 8x_2 - x_3 &= 35 \end{aligned} \quad (3)$$

5A) Find all the eigen values and any two of the eigen vectors of the matrix

$$A = \begin{bmatrix} 4 & 2 & -2 \\ -5 & 3 & 2 \\ -2 & 4 & 1 \end{bmatrix}. \quad (4)$$

5B) Solve by method of variation of parameters: $\frac{d^2y}{dx^2} + 4y = 4 \sec^2 2x$ (3)

5C) Using Gauss-Seidel method, solve the system of linear equations:

$$x + y + 54z = 110$$

$$27x + 6y - z = 54$$

$$6x + 15y + 2z = 72 \quad (3)$$

Perform 3 iterations and correct the solution up to four decimal places by taking the initial approximation $x^{(0)} = y^{(0)} = z^{(0)} = 0$.

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